

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

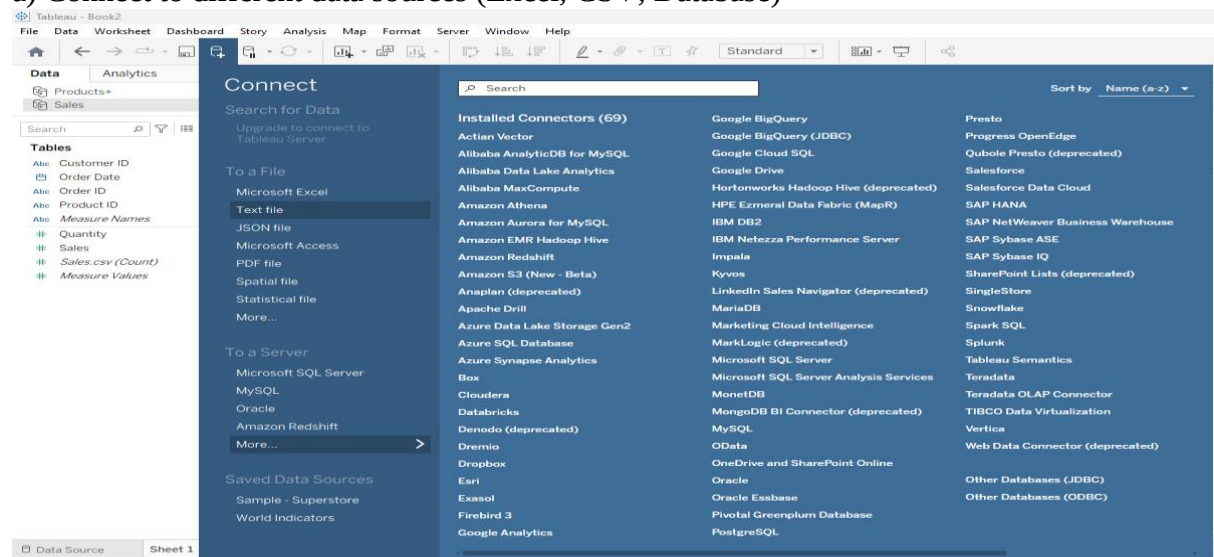
Practical – II

Roll No.: T003	Name: Pranay Bhadwalkar
Class: TYIT	Batch: 01
Date of Assignment: 14/08/2026	Date/Time of Submission: 14/08/2026

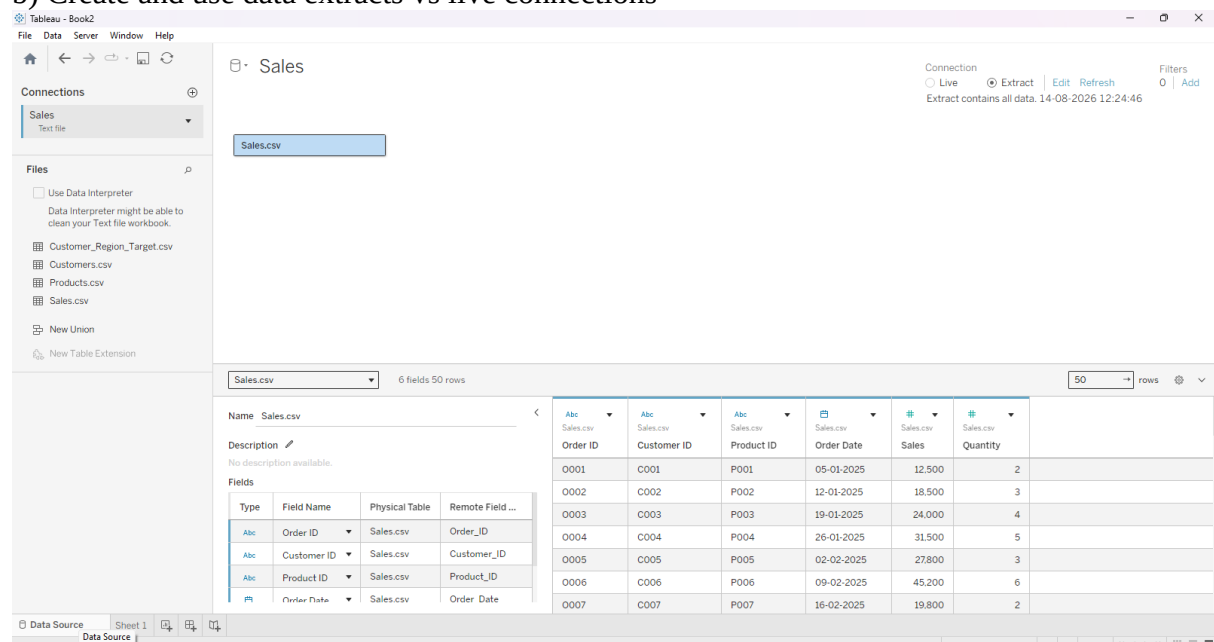
Part B – Program

Working with Data Sources and Data Preparation in Tableau

a) Connect to different data sources (Excel, CSV, Database)



b) Create and use data extracts vs live connections



c) Manage metadata (rename, hide, change data types)

Connection: ☐ Live ☒ Extract | Ed
Extract contains all data. 14-01-2025

Sales.csv

6 fields 50 rows

Name: Sales.csv
Description: No description available.

Type	Field Name	Physical Table	Remote Field ...
Abc	Order ID	Sales.csv	Order_ID
Abc	Customer ID	Sales.csv	Customer_ID
Abc	Product ID	Sales.csv	Product_ID
Abc	Order Date	Sales.csv	Order Date

Order ID	Customer ID	Product ID	Order Date	Sales	Quantity
0001	C001	P001	05-01-2025	12,500	2
0002	C002	P002	12-01-2025	18,500	3
0003	C003	P003	19-01-2025	24,000	4
0004	C004	P004	26-01-2025	31,500	5
0005	C005	P005	02-02-2025	27,800	3
0006	C006	P006	09-02-2025	45,200	6
0007	C007	P007	16-02-2025	19,800	2

Tableau - Book2

File Data Worksheet Dashboard Story Analysis Map Format Server Window

Data Analytics

Products+ Sales

Search

Tables

- Abc Customer ID
- Abc Order Date
- Abc Order ID
- Abc Product ID
- Abc Measure Name
- # Quantity
- # Sales
- # Sales.csv (C)
- # Measure Value

Pages

Filters

Marks

Sheet 1

Columns

Rows

Change Data Type

- Number (decimal)
- Number (whole)
- Date & Time
- Date
- String
- Default

d) Perform joins (inner, left, right) on multiple tables

inner join :

Products+ Connection ☐ Live ☒ Extract Edit Refresh Filters 0 | Add
Extract will contain all data.

Products.csv is made of 2 tables. ⓘ

Products.csv Sales.csv1

Join

Inner Left Right Full Outer

Data Source Sales.csv1

Product ID Product ID (Sales.csv1)

Add new join clause

Products.csv 11 fields 250 rows 100 rows

Name Products.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remo...
Abc	Product ID	Products.csv	Produc...
Abc	Product Name	Products.csv	Produc...
Abc	Category	Products.csv	Catego...
Abc	Brand	Products.csv	Brand

Abc	Abc	Abc	Abc	#	Abc	Abc
Products.csv	Products.csv	Products.csv	Products.csv	Products.csv	Sales.csv1	Sales.csv1
Product ID	Product Name	Category	Brand	Unit Price	Order ID (Sales.csv1)	Customer ID (Sales.csv1)
P001	Laptop	Electronics	Dell	55,000	O041	C041
P001	Laptop	Electronics	Dell	55,000	O031	C031
P001	Laptop	Electronics	Dell	55,000	O021	C021
P001	Laptop	Electronics	Dell	55,000	O011	C011
P001	Laptop	Electronics	Dell	55,000	O001	C001
P002	Mobile	Electronics	Samsung	25,000	O042	C042
P002	Mobile	Electronics	Samsung	25,000	O032	C032

left join :

Products+ Connection ☐ Live ☒ Extract Edit Refresh Filters 0 | Add
Extract will contain all data.

Products.csv is made of 2 tables. ⓘ

Products.csv Sales.csv1

Join

Inner Left Right Full Outer

Data Source Sales.csv1

Product ID Product ID (Sales.csv1)

Add new join clause

Products.csv 11 fields 250 rows 100 rows

Name Products.csv

Description No description available.

Fields

Type	Field Name	Physical Table	Remo...
Abc	Product ID	Products.csv	Produc...
Abc	Product Name	Products.csv	Produc...
Abc	Category	Products.csv	Catego...
Abc	Brand	Products.csv	Brand

Abc	Abc	Abc	Abc	#	Abc	Abc
Products.csv	Products.csv	Products.csv	Products.csv	Products.csv	Sales.csv1	Sales.csv1
Product ID	Products.csv.Product ID	Category	Brand	Unit Price	Order ID (Sales.csv1)	Customer ID (Sales.csv1)
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P002	Mobile	Electronics	Samsung	25,000	O002	C002
P002	Mobile	Electronics	Samsung	25,000	O002	C002

right join :

Products+ Connection ☐ Live ☒ Extract Edit Refresh Filters 0 | Add
Extract will contain all data.

Products.csv is made of 2 tables. ⓘ

Products.csv Sales.csv1

Join

Inner Left Right Full Outer

Data Source Sales.csv1

Product ID Product ID (Sales.csv1)

Add new join clause

Products.csv 11 fields 250 rows 100 rows

Name Products.csv

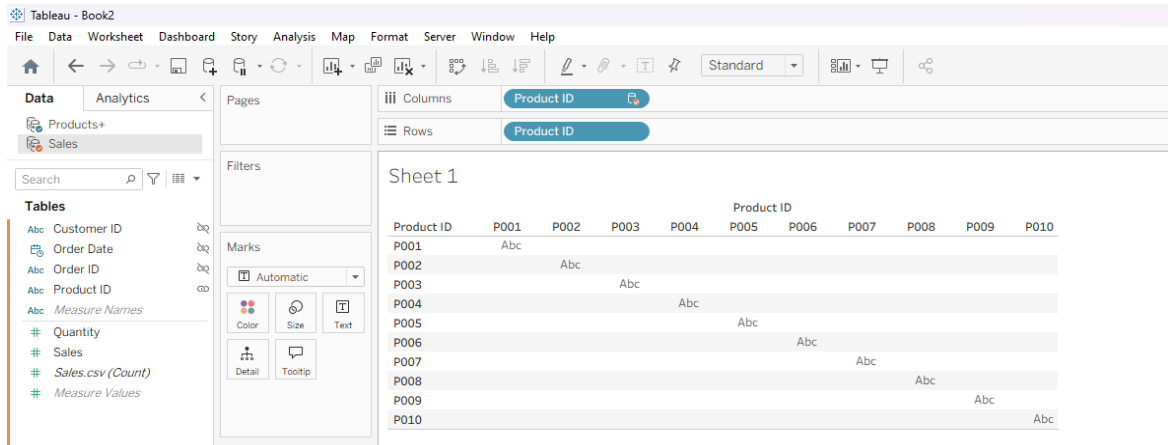
Description No description available.

Fields

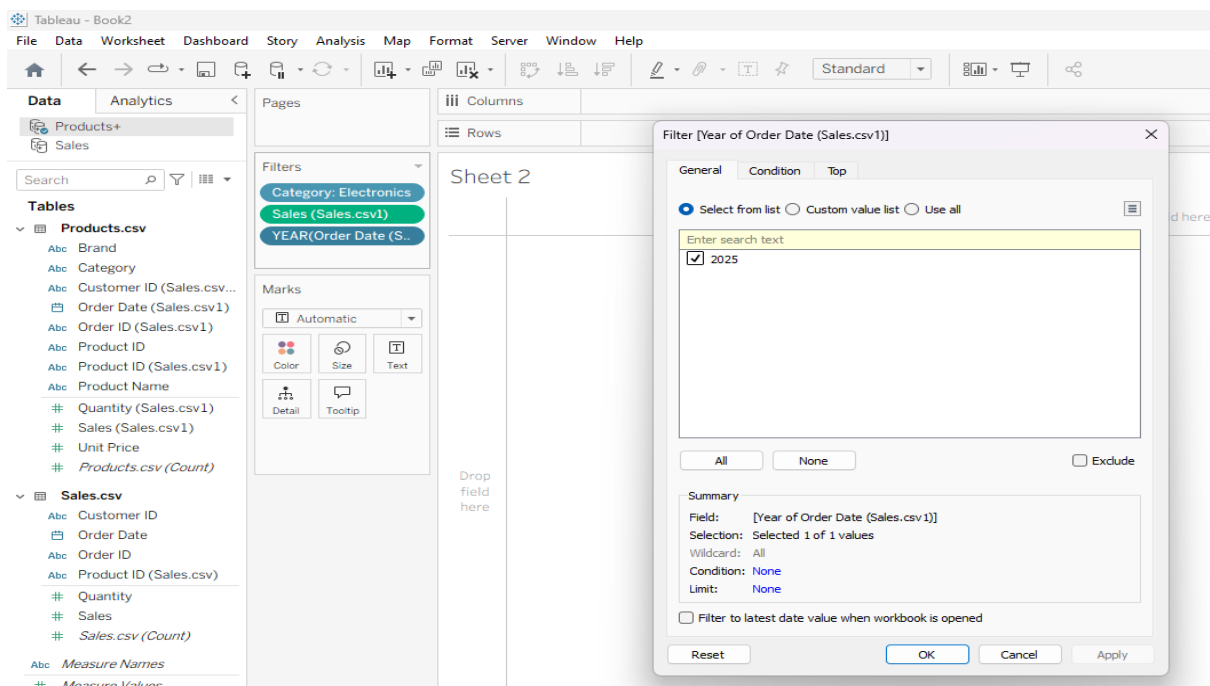
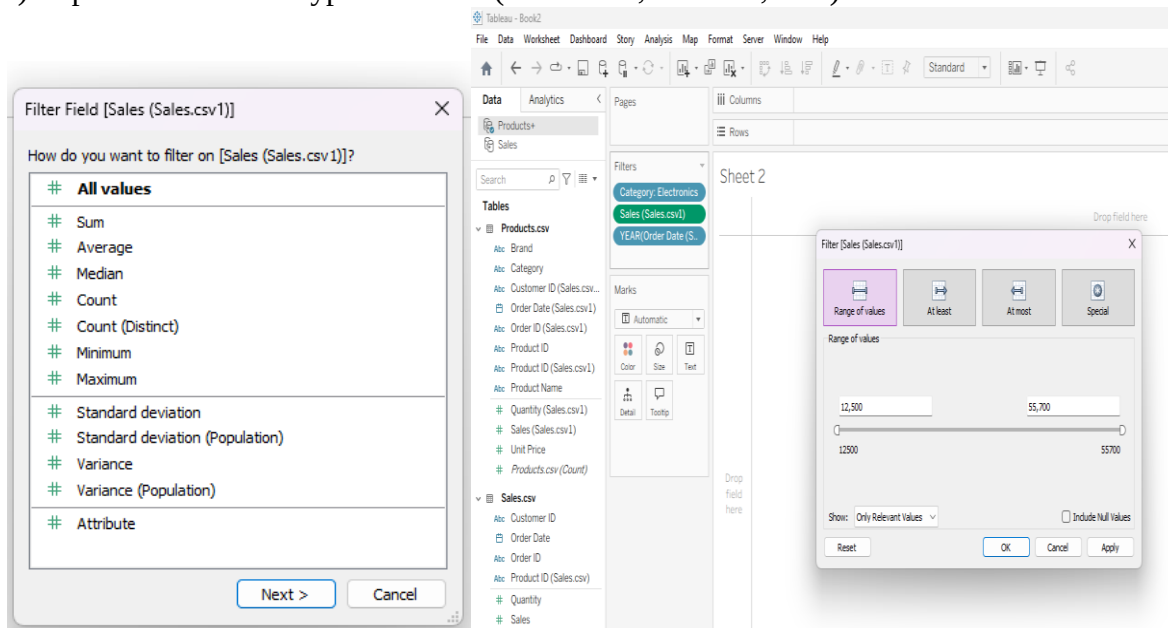
Type	Field Name	Physical Table	Remo...
Abc	Product ID	Products.csv	Produc...
Abc	Product Name	Products.csv	Produc...
Abc	Category	Products.csv	Catego...
Abc	Brand	Products.csv	Brand

Abc	Abc	Abc	Abc	#	Abc	Abc
Products.csv	Products.csv	Products.csv	Products.csv	Products.csv	Sales.csv1	Sales.csv1
Product ID	Product Name	Category	Brand	Unit Price	Order ID (Sales.csv1)	Customer ID (Sales.csv1)
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P001	Laptop	Electronics	Dell	55,000	O001	C001
P002	Mobile	Electronics	Samsung	25,000	O002	C002
P002	Mobile	Electronics	Samsung	25,000	O002	C002

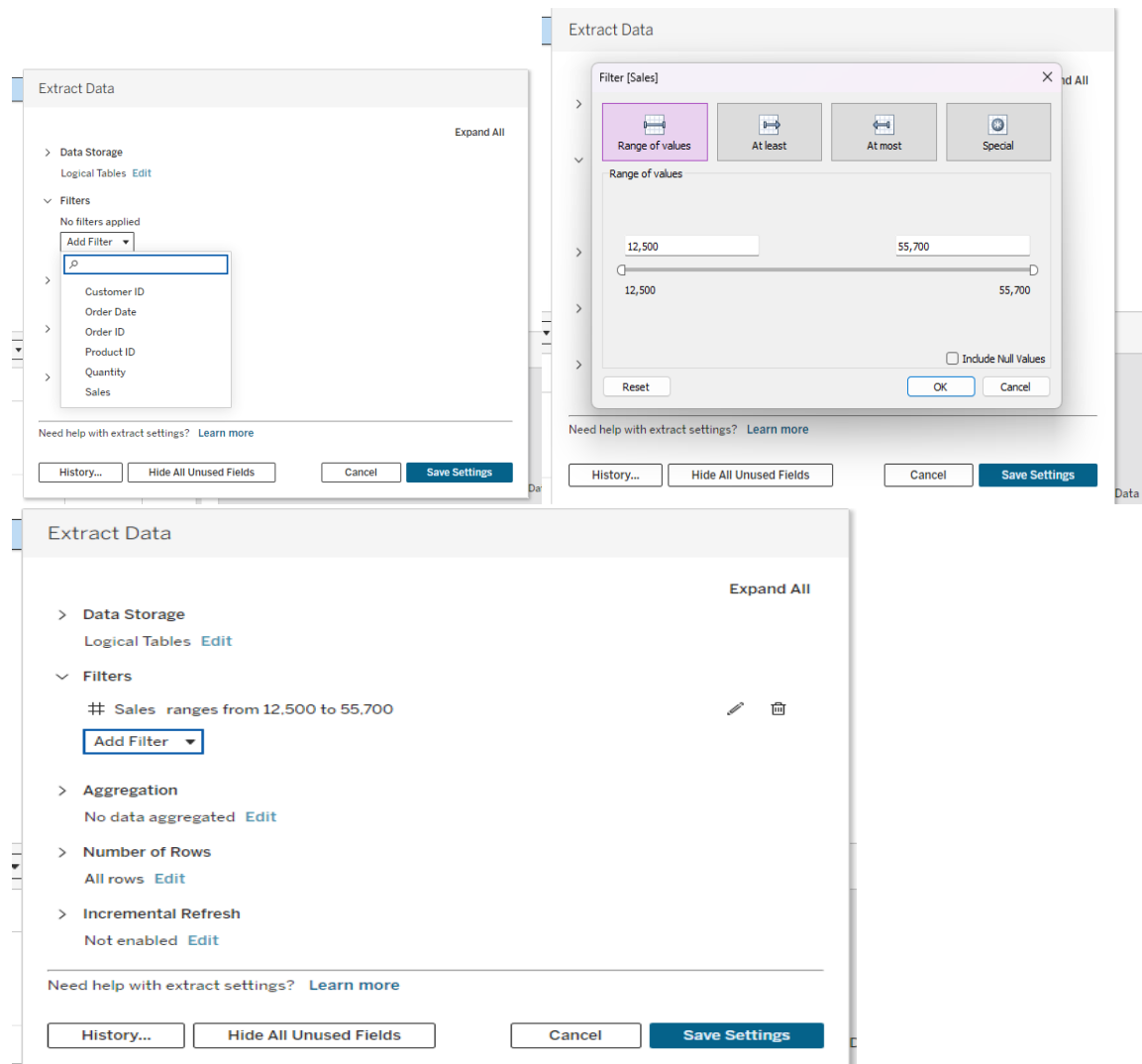
e) Apply data blending between two sources



f) Implement different types of filters (dimension, measure, date)



g) Optimize performance using extracts and filtering



dashboard :

Dashboard | Layout <

Default
Phone
Device Preview

Size
Desktop Browser (1000 x 8... >

Sheets
Sheet 1
Sheet 2

Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Blank
Workflow
Web Page
Tiled Floating
Show dashboard title

Practical II – Data Sources and Data Preparation in Tableau
Name: Pranay Bhadwalkar
Roll No.: T003

Sheet 1

Product ID	P001	P002	P003	P004	P005	P006
P001	Abc					
P002		Abc				
P003			Abc			
P004				Abc		
P005					Abc	
P006						
P007						
P008						
P009						
P010						

Sheet 2

Brand	Ye..
Canon	2..
Dell	
Lenovo	
LG	
Samsung	

Quantity (Sales.csv1)

50
60
80
100
120
140
160
170